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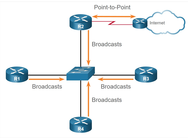
Multiaccess OSPF Networks

OSPF Network Types

Another type of network that runs OSPF is the multiaccess OSPF network. Multiaccess OSPF networks are unique in that one router controls the distribution of LSUPs. The router that is selected for this role should be determined by the network administrator through proper configuration.

Routers can be connected to networks possesses depending on the type of network. The previous topology used point-to-point links between the routers. However routers can be connected to the same switch to form a multiaccess network, as shown in the topology. The OSPF protocol will automatically establish neighbor relationships. In broadcast networks, all devices on the network see all broadcast and multicast traffic.

OSPF Multiaccess Network



OSPF Designated Router

Recall that, in multiaccess networks, OSPF elects a DR and BDR as a solution to manage the number of adjacencies and the flooding of link-state advertisements (LSAs). The DR and BDR extend the operation of the Hello protocol. Distributing LSAs and exchange The DR uses the multicast IPv4 address 224.0.0.5 which is known to support all OSPF routers.

A BDR is also elected in case the DR fails. The BDR listens passively and maintains a relationship with all the routers. If the DR stops producing Hello packets, the BDR promotes itself and assumes the role of DR.

All other routers become a DROTHER (in router that is neither the DR nor the BDR). DROTHERs use the multicast address 224.0.0.6 (all designated routers) to send OSPF packets to the DR and BDR. Only the DR and BDR listen for 224.0.0.6.

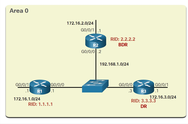
In the figure, R1, R3, and R4 are DROTHERs. Once play to see the animation of R2 acting as DR. Notice that only the DR and the BDR process the LSA sent by R1 using the multicast IPv4 address 224.0.0.6. The DR then sends out the LSA to all OSPF routers using the multicast IPv4 address 224.0.0.5.

Role of the DR

OSPF Multiaccess Reference Topology

In the multiaccess topology shown in the figure, there are three routers interconnected over a common Ethernet multiaccess network, 192.168.1.0/24. Each router is configured with the indicated IPv4 addresses on the Gigabit Ethernet 0/0/0 interface.

Because the routers are connected over a common multiaccess network, OSPF has automatically elected a DR and BDR. In this example, R2 has been elected as the DR because its router ID is 2.2.2.2, which is the highest in this network. R3 is the BDR because it has the second highest router ID in the network.



Verify OSPF Router Roles

To verify the roles of the OSPFv2 router, use the **show ip ospf interface** command.

Click each button see the output for the show ip ospf interface command on each router.

R1 (DROTHER) | R2 BDR | R3 DR

R1 DROTHER

The output generated by R1 confirms that the following:

1. R1 is not the DR or BDR, but is a DROTHER with a default priority of 1. (Line 1)
2. The router ID for the DR is 3.3.3.3 and IPv4 address 192.168.1.3, while the BDR is R2 with router ID 2.2.2.2 and IPv4 address 192.168.1.2. (Lines 4 and 5)
3. R1 has two adjacencies: one with the BDR and one with the DR. (Lines 20-22)

```
R1#show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 1
    No priority      1      3      Disabled      Disabled      Disabled
    Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
    Index 1/1/1, flood queue length 0
    Last flood scan length is 1, max is 2
    Last flood scan time is 0 msec, max is 1 msec
    Neighbor Count is 2, Adjacent neighbor count is 2
      Adjacent with neighbor 2.2.2.2  (Designated Router)
      Adjacent with neighbor 3.3.3.3  (Backup Designated Router)
    Suppress hello for 0 neighbor(s)
R1#
```

Verify DR/BDR Adjacencies

To verify the OSPFv2 adjacencies, use the **show ip ospf neighbor** command, as shown in the example for R1. The state of neighbors in multiaccess networks can be as follows:

- **FULL/DROTHER** - This is a DR or BDR router that is fully adjacent with a non-DR or BDR router. These two neighbors can exchange Hello packets, updates, queries, replies, and acknowledgments.
- **FULL/BDR** - This is a DR router that is fully adjacent with a backup DR router. These two neighbors can exchange Hello packets, updates, queries, replies, and acknowledgments.
- **FULL/DR** - The router is fully adjacent with the indicated BDR neighbor. These two neighbors can exchange Hello packets, updates, queries, replies, and acknowledgments.
- **2-WAY/DROTHER** - The router DR or BDR router has a neighbor relationship with another non-DR or BDR router. These two neighbors exchange Hello packets.

The normal state for an OSPF router is usually FULL. If a router is stuck in another state, it is an indication that there are problems in forming adjacencies. The only exception to this is the 2-WAY state, which is normal in a multiaccess broadcast network. Multiaccess networks elect a DR and a BDR. All routers establish adjacency with any DROTHERs that join the network. When this happens, the neighbor state displays as 2-WAY/DROTHER.

Click each button see the output for the show ip ospf neighbor command on each router.

R1 (DROTHER) | R1 Adjacencies | R2 Adjacencies

The output generated by R1 confirms that R1 has adjacencies with the following routers:

- R2 with router ID 2.2.2.2 (it is in a Full state and the role of R2 is BDR)
- R3 with router ID 3.3.3.3 (it is in a Full state and the role of R3 is DR)

```
R1#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:35    192.168.1.2     GigabitEthernet0/0/0
3.3.3.3           1   FULL/DR         00:00:35    192.168.1.3     GigabitEthernet0/0/0
R1#
```

Default DR/BDR Election Process

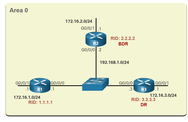
How do the DR and BDR get elected? The OSPF DR and BDR election decision is based on the following criteria, in sequential order:

1. The routers in the network elect the router with the highest interface priority as the DR. The router with the second highest interface priority is elected as the BDR. The default priority for any router interface is 1. (RID: If the interface priority value is set to 0, that interface cannot be elected as DR nor BDR. The default priority of multiaccess broadcast interfaces is 1. If the router interfaces are configured with routers have an equal priority value and must rely on another tie-breaking method during the DROTHER election.
2. If the interface priorities are equal, then the router with the highest router ID is elected the DR. The router with the second highest router ID is the BDR.

Recall that the router ID is determined in one of the following three ways:

1. The router ID can be manually configured.
2. If no router ID is are configured, the router ID is determined by the highest loopback IPv4 address.
3. If no loopback interfaces are configured, the router ID is determined by the highest active IPv4 address.

OSPFv2 Multiaccess Reference Topology



In the figure, all Ethernet router interfaces have a default priority of 1. As a result, based on the selection criteria listed above, the router with the highest router ID, R3 and R3, R3 with the highest router ID becomes the DR, and R2, with the second highest router ID, becomes the BDR.

The DR and BDR election process takes place as soon as the first router with an OSPF-enabled interface is active on the multiaccess network. This can happen when the preconfigured OSPF routers are powered on, or when OSPF is activated on the interface. The election process only takes a few seconds. If all of the routers on the multiaccess network have not finished booting, it is possible that a router with a lower router ID becomes the DR.

OSPF DR and BDR elections are not pre-emptive. If a new router with a higher priority or higher router ID is added to the network after the DR and BDR have been elected, the existing DR and BDR remain the DR and BDR. This is because these roles have already been assigned. The addition of a new router does not initiate a new election process.

DR Failure and Recovery

After the DR is elected, it remains the DR until one of the following events occurs:

- The DR fail.
- The OSPF process on the DR fails or is stopped.
- The multiaccess interface on the DR fails or is shutdown.

If the DR fails, the BDR is automatically promoted to DR. This is the case even if another DROTHER with a higher priority or router ID is added to the network after the initial DR election. However, after a BDR is promoted to DR, a new BDR election occurs and the DROTHER with the highest priority or router ID is elected as the new BDR.

Click each button for an illustration of various scenarios relating to the DR and BDR election process.

R1 Fails | R2 No Longer Network | R4 Joins Network | R2 Fails

In this scenario, the current DR (R3) fails. Therefore, the pre-elected BDR (R2) assumes the role of DR. Subsequently, an election is held to choose a new BDR. Because R1 is the only DROTHER, it is elected as the BDR.

The ip ospf priority Command

If the interface priorities are equal on all routers, the router with the highest router ID is elected the DR. It is possible to configure the router ID on a router to ensure that it is not the DR or BDR router, but this router will then have to be configured to have the router ID on all routers. Configuring the router ID in order to help control this. However, in large networks this can be cumbersome.

Instead of relying on the router ID, it is better to control the election by setting interface priorities. This also allows a router to be the DR on one network and a DROTHER on another network. To set the priority on an interface, use the command **ip ospf priority** value, where value is 0 to 255. A value of 0 does not become a DR or a BDR. A value of 1 to 255 on the interface makes it more likely that the router becomes the DR or the BDR.

Configure OSPF Priority

In the topology, the **ip ospf priority** command will be used to change the DR and BDR as follows:

- R1 should be the DR and will be configured with a priority of 255.
- R2 should be the BDR and will be left with the default priority of 1.
- R3 should never be a DR or BDR and will be configured with a priority of 0.

Change the R1 (R3-R2) interface priority from 1 to 255.

```
R1(config)# interface GigabitEthernet 0/0/0
R1(config-if)# ip ospf priority 255
R1(config-if)# end
```

Change the R3 (R2-R1) interface priority from 1 to 0.

```
R3(config)# interface GigabitEthernet 0/0/0
R3(config-if)# ip ospf priority 0
R3(config-if)# end
```

The following example, shows how to clear the OSPF process on R1. The **clear ip ospf process** command also must be entered on R2 and R3 (not shown). Notice the OSPF state information that is generated.

```
R1#clear ip ospf process
Reset ALL OSPF processes? [no]: y
R1#
*Mar  1 00:27:41.501: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from FULL to DOWN,
Neighbor Down: Interface down or detached
*Mar  1 00:27:41.501: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/0 from FULL to DOWN,
Neighbor Down: Interface down or detached
*Mar  1 00:27:41.501: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from LOADING to
FULL, Loading Done
*Mar  1 00:27:41.501: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/0 from LOADING to
FULL, Loading Done
```

The output from the **show ip ospf interface g0/0/0** command on R1 confirms that R1 is now the DR with a priority of 255 and identifies the new neighbor adjacencies of R2.

```
R1#show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 255
    No priority      1      3      Disabled      Disabled      Disabled
    Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
    Index 1/1/1, flood queue length 0
    Last flood scan length is 1, max is 2
    Last flood scan time is 0 msec, max is 1 msec
    Neighbor Count is 2, Adjacent neighbor count is 2
      Adjacent with neighbor 2.2.2.2  (Backup Designated Router)
      Adjacent with neighbor 3.3.3.3
    Suppress hello for 0 neighbor(s)
R1#
```

Syntax Checker - Configure OSPF Priority

Use the Syntax Checker to configure a different OSPF priority scenario for R1, R2, and R3.

Use the **ip ospf priority** command to change the DR and BDR as follows:

- R1 should be the BDR and will be configured with a priority of 10.
- R2 should never be a DR or BDR and will be configured with a priority of 0.
- R3 should be the DR and will be left with its default priority of 100.

On all routers, use **g0/0/0** for the interface name.

You are logged into R1 in global configuration mode. Configure R1 with a priority of 10.

```
R1(config)# interface g0/0/0
R1(config)#
```

Reset | Show Me | Show All

Packet Tracer - Determine the DR and BDR

In this activity, you will complete the following:

- Examine DR and BDR roles and watch the roles change when there is a change in the network.
- Modify the priority to control the roles and force a new election.
- Verify routers are filling the desired roles.

1. Determine the DR and BDR
2. Determine the DR and BDR